

Agentic AI Course Planner

Problem statement and motivation:

Currently, Georgia state university has a shortage of advisement services available to students. Advisors are overloaded, and are often limited to 15 minute sessions with students to plan out multiple semesters. While GSU services like Navigate and DegreeWorks are available to students for self-advisement, it can be difficult to create a long term plan using these tools. With an agentic AI system, students will be able to create a multi-semester course plan by themselves. This solution would provide students an intuitive way to plan their own course of study. Advisors would also be able to maximize efficiency during advisement meetings, as students will be prepared with a plan for what they want to do. Through this system, the user would be able to generate and update multi-semester plans and compare them without needing to contact an advisor.

Use Cases:

The AI's connection to the GSU course catalog would allow it to generate specialized programs of study for each individual student based on their interests. For example, a student may be majoring in cybersecurity, but interested in taking healthcare electives. The agent would be able to search for suitable electives within the GSU course catalog. It would also check to ensure that the student meets the CPOS requirements for an elective. This way, the student is able to graduate with specialized domain knowledge that gives them an edge in related fields, such as healthcare information technology.

Should the requirements for a degree change, this information would be automatically added to the AI's database. This would be immediately reflected in the student's advisement plan due to the RAG output style of the agent. Additionally, this would spare the human advisors from having to redo everyone's academic evaluation by hand.

A student would also be able to generate multiple different course plans based on different interests. While the current DegreeWorks system does allow students to generate "What-If" scenarios to view required classes, it does not advise students on when to take classes. Students must rely on meeting with advisors to plan out multiple semesters, especially when taking on additional programs like the honors college or research laureates. This AI system would be able to take natural language requests such as "What courses should I take for the next 3 years to graduate with a degree in CIS, an honors laureate, and a concentration in data analytics? I also want to learn more about financial management." and create a schedule of appropriate classes and electives for the student. This way, the user can easily create multiple what-if schedules through prompting the system.

Initial system design:

Agents and roles:

- Catalog agent: Chooses courses based on the student's request
- Ordering agent: Orders the courses based on prerequisites
- Ranking agent: Ranks the difficulty and necessity of each class to create a balanced schedule
- Planner: attaches all of it to a calendar

- Propositional logic agent: Checks the logic of the created schedule to make sure everything meets CPOS requirements
- Advisor Agent: Creates a report explaining the created schedule to the student

Expected tools/ API sources:

- GSU course catalog
- GSU semester calendars
- The student's current academic evaluation
- List of degree requirements from the department
- List of registration rules (ex: a student can't take more than 18 credit hours at a time)
- Calendar interface with the ability to update it

Feasibility and Risks:

This agent would rely heavily on the provided rules. If the requirements for a degree change and the AI's resources are not updated, it would have a cascading effect on the way that it creates course schedules. Additionally, hallucination or poor output from this system could have serious financial and academic repercussions on the user. Finally, the AI must also use private information like the student's personal academic evaluation. A simpler version could solely work with public course catalogs and degree requirements that GSU has uploaded to their website. The system should be accessing the site's information legally.